

**System and Complexity Science:
The Generation, Persistence, and Evolution of Order**

A Physics Constitution for Open Complex Giant Systems

Grit Meng (Meng Fanchun)

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Preface: Practice Origin

From 2007 to 2020, I worked at Lenovo Group. During those thirteen years, my team and I accomplished one monumental task: on the world’s most intricate discrete manufacturing battlefield, we built an end-to-end closed-loop autonomous decision-making supply chain planning system, known internally as IPS (“Integrated Planning Solution”).

This system represents one of the few engineering verifications in human history that achieved a “computable closed-loop” for governing open complex giant systems: autonomous decision-making with Human-in-the-Loop-Out (HILO). Under its scheduling, delivery response rate surged from 54% to 98%, order accuracy increased by 32%, and inventory turnover improved by 1.9x, releasing billions of RMB in liquidity.

This success forced me to pursue the fundamental question: Why did it work when traditional software giants failed?

Traditional software giants were trapped in the reductionist paradigm, treating non-linear networks as linear processes. Through engineering verification, we proved the power of Prof. Qian Xuesen’s “Metasynthetic Engineering” and “Human-Machine Integration” in thousand-billion-level physical battlefields. This led to my formulation of the foundational physics for complex system governance, culminating in *Holographic Anti-Entropy: A Physics Constitution for Open Complex Giant Systems*.

We did not bypass the two ultimate puzzles of complexity science—“Emergence” and “Computational Irreducibility”—but confronted them with formal proofs, establishing a universal algorithmic path for computable science.

Experience, philosophy, religion, and science are all systems of order constructed by observer boundary definition. This monograph presents the first-principles physics of order itself: how order emerges from chaos, persists against entropy, and evolves across generations.

Introduction: The Origin of Order and Meta-Cognitive Algorithms

Human exploration of the world over thousands of years is fundamentally the process of an observer defining boundaries to construct order.

This monograph, *System and Complexity Science: The Generation, Persistence, and Evolution of Order*, steps back to the absolute logical origin. Using Occam's Razor to strip away intermediate redundancies, it establishes the first-principles physics of order generation, persistence, and evolution at the dimension of meta-cognitive algorithms: boundary definition creates order, rigid constraints persist order, and residual feedback evolves order.

Part I

Volume I: Order — The Origin of Order: First Principles of Open Complex Giant Systems

Chapter 1

Definitions and Axioms

1.1 The Observer and Boundary Definition (Definition 1)

Without an observer, there is no world. Once observed, chaos immediately manifests as the world. Above physics, the world is order.

Definition 1 (Disorder): The intrinsic state of being prior to observer boundary definition is termed Disorder. It is an undivided whole.

Definition 2 (Motion): Within disorder, there is intrinsic motion. Motion arises spontaneously, subsides spontaneously, and repeats endlessly.

Definition 3 (Observer, Wisdom, and Action): When a localized region maintains its distinction from its environment, it forms an Observer. The observer possesses two coexisting dimensions: Wisdom (*Zhi*, the capacity for reflection) and Action (*Shi*, the capacity for construction).

Axiom 0 (Zeroth Law – Observer Emergence Law): The observer is not a pre-given starting point, but a persistent steady state naturally reached when internal motion achieves critical density. Without an observer, there is no world.

Definition 4 (Boundary Definition and Order Construction): The observer's act of partitioning boundaries via *Zhi* is termed Boundary Definition (Π); establishing internal rules via *Shi* is termed Order Construction (Ω).

1.2 Real, Ideal, and Residual (Definition 2)

Simultaneous with boundary definition, three layers of reality manifest:

- **Real (Ξ):** The infinite, silent background of physical reality.
- **Ideal (Ω):** The localized, structured framework established by the observer inside the boundary.
- **Residual (Δ):** The discrepancy interface computed between the real observation and ideal framework. Residual Δ acts as the sole driver for evolutionary paradigm shifts.

1.3 The Three First-Principles Axioms of Order

Axiom 1 (First Law – Generation Axiom / Collapse Without Ideal Framework): Without observer boundary definition establishing an ideal framework Ω , everything within chaos collapses into void.

Axiom 2 (Second Law – Persistence Axiom / Dissipation Without Rigid Constraints): Without rigid constraints and self-consistent rules $\Pi_{\perp}$, elements inside the framework dissipate due to uncoordinated friction.

Axiom 3 (Third Law – Evolution Axiom / Stagnation Without Residual Feedback): Without identifying residual Δ , the framework solidifies into dogma and generational evolution halts.

Chapter 2

The Holographic Meta-Theorem and Dynamic Essence

2.1 The Holographic Meta-Theorem

Order persists and evolves across generations if and only if it simultaneously satisfies three necessary and sufficient conditions:

$$\text{Meta-Theorem} \iff \text{Boundary Definition } \mathbf{\Pi} \implies \text{Prevent Collapse} \quad \wedge \quad \text{Rigid Constraint } \mathbf{\Pi}_{\perp} \implies \text{Prevent} \quad (2.1)$$

2.2 The Dynamic Essence of Order and Human Constructs

Order is never a static objective entity. Its physical and philosophical essence lies in the dynamic closed loop of boundary definition creating order, rigid constraints persisting order, and residual feedback evolving order.

Chapter 3

Formal Preview and Falsifiability Boundaries

3.1 Three Falsifiability Deadlines

This system declares its absolute falsifiability deadlines:

1. Order persists without an ideal framework Ω ;
2. Coordinated collaboration occurs without rigid constraints $\Pi_{\perp}$;
3. Generational evolution continues while residual Δ remains identically zero.

Chapter 4

Isomorphism: Continuing the Unfinished Inquiries of Predecessors

The boundary-ideal-residual closed loop connects the unfinished inquiries of history's greatest minds:

1. **Laozi & Zhuangzi:** "Nameless is the origin of Heaven and Earth" (Disorder) $\rightarrow$ "Named is the mother of all things" (**II**).
2. **Yogacara Buddhism:** "Three realms are mind-only, all phenomena are consciousness-only" $\rightarrow$ Observer boundary definition.
3. **Wang Yangming:** "No object exists outside the mind" and "Conscience" (**E_{sp}**).
4. **Immanuel Kant:** "Understanding prescribes laws to nature."
5. **G. Spencer-Brown:** *Laws of Form* – "Draw a distinction."
6. **John A. Wheeler:** Quantum Physics – "It from Bit."
7. **Erwin Schrödinger & Karl Friston:** Negative Entropy and Free Energy Principle.
8. **Norbert Wiener & W. Ross Ashby:** Cybernetics, Feedback, and Law of Requisite Variety.
9. **Stuart Kauffman & Herbert Simon:** Edge of Chaos and Bounded Rationality.
10. **Karl Popper:** Falsifiability driven by counterexamples.
11. **Pansystems Theory (Prof. Xuemou Wu):** Mathematical formalization of system boundaries, transformations, and relation networks.
12. **Non-IID Data Dynamics (Prof. Longbing Cao):** Non-independent and identically distributed dynamics and factorial complexity $O(N!)$ emergence.
13. **Prof. Qian Xuesen (H. S. Tsien):** Open Complex Giant Systems, Metasynthetic Engineering, and Human-Machine Integration.

Part II

Volume II: Formal Work Operators and Physics Constitution — Ideal Evolution, Paradigm Shift, and Work Operators

Chapter 5

Ideal World Evolution and the 400-Year Scientific Dilemma

5.1 The Physical Essence of the Ideal World

The ideal world Ω is a localized negative entropy projection built inside unpartitioned physical reality Ξ .

5.2 Reductionism's Degenerate Limit and Factorial Complexity $O(N!)$ Emergence

Reductionism is the degenerate limit of complex systems in simple, single-element states ($N \rightarrow 1$). When elements scale and exhibit non-linear coupling and gaming, the system undergoes a spontaneous phase transition, emerging into factorial complexity:

$$\text{Complexity} \sim O(N!) \tag{5.1}$$

Chapter 6

Paradigm Breakthrough: Thousand-Billion Empirical Proof and Computable Work

6.1 Confronting Emergence and Computational Irreducibility

Through 13 years of engineering practice in global discrete manufacturing, we demonstrated that open complex giant systems possess a computable closed loop by executing algebraic pruning via operator Π and driving evolution via residual Δ .

Chapter 7

Formal Work Operators and Topological Manifolds

7.1 The A Priori Boundary Operator Π

The projection operator Π maps infinite phase space Ξ to finite state space Ω :

$$\Pi^2 = \Pi, \quad \dim(\Omega) \ll \dim(\Xi) \quad (7.1)$$

7.2 Rigid Manifold $\Pi_{\perp}$ and Persistence Work

State trajectory $\mathbf{x}(t)$ is constrained upon rigid manifold $\Pi_{\perp}$:

$$\mathbf{x}(t) \in \Pi_{\perp}, \quad \mathbf{x}_{\perp}(t) \rightarrow 0 \quad (7.2)$$

7.3 Residual Difference Δ and Second-Order Meta-Cognitive Operator Φ

When residual $\Delta > \theta_{\text{trigger}}$, operator Φ rewrites the orthogonal basis:

$$\Phi : \Pi_k \longrightarrow \Pi_{k+1} \quad (7.3)$$

Chapter 8

Operator Embodiment: 5D Topological Mind and Neural Mapping

8.1 1:1 Topological Mapping

Operators map directly to brain neural circuits: Conscience ($\mathbf{E}_{\text{sp}}$ / DMN), Hypersensitivity (AMY-LC-NE), Affect (SN), Fluid Intelligence (FPN), and Meta-cognition (Φ / rPFC BA10 area).

8.2 Conscience Operator $\mathbf{E}_{\text{sp}}$ and Goodhart Collapse Truncation

When optimization pressure approaches infinity, Goodhart's Law causes expectation utility to diverge to negative infinity. Conscience operator $\mathbf{E}_{\text{sp}}$ truncates heavy-tailed distribution $p(\Delta)$ into sub-Gaussian distribution:

$$\mathbf{E}_{\text{sp}} \cdot p(\Delta) \in \text{Sub-Gaussian} \quad (8.1)$$

8.3 Resolution of Gödel's Incompleteness and Turing Halting

Human rPFC (BA10 area) expanded 4.7x in evolution, acting as the physical entity of second-order operator Φ . Under deadlock, Φ computes basis rescaling within reset time:

$$\Delta t_{\text{rewrite}} \leq \frac{S_{\text{ctrl}}}{k_B \ln 2 \cdot \|\mathbf{\Pi}_{k+1} - \mathbf{\Pi}_k\|} \quad (8.2)$$

Part III

Volume III: Physics Constitution and Dimensionality Reduction Codex — Dimensional Projection of System Engineering

Chapter 9

Teleology: Three Physical Pillars of Holographic Anti-Entropy

9.1 Mismatch Angle θ and Effective Work Conversion

The mismatch angle θ between intention vector $\mathbf{V}_{\text{int}}$ and logical trajectory $\mathbf{V}_{\text{logic}}$ dictates work conversion:

$$W_{\text{eff}} = W_{\text{total}} \cdot \cos \theta \quad (9.1)$$

Boundary operator $\mathbf{\Pi}$ and rigid manifold $\mathbf{\Pi}_{\perp}$ force $\cos \theta \rightarrow 1$.

9.2 Control Sampling and Feedback Time Window Δt

Control sampling and self-healing closed-loop periods are strictly bounded by three physical laws:

1. **Nyquist-Shannon Control Criterion:** $\Delta t_{\text{sense}} \leq \frac{1}{2f_{\text{max}}}$.
2. **Lyapunov Prediction Window:** $\Delta t_{\text{heal}} \leq \frac{1}{\lambda_{\text{max}}}$ (enforcing millisecond IPS self-healing).
3. **Second-Order Paradigm Reset Window:** $\Delta t_{\text{rewrite}} \leq \frac{S_{\text{ctrl}}}{k_B \ln 2 \cdot \|\mathbf{\Pi}_{k+1} - \mathbf{\Pi}_k\|}$.

Chapter 10

Ontology: From Single Point Reduction to HILO Symbiosis

Abandoning single-point carbon intervention to achieve "Human-in-the-Loop-Out, Human-Machine Symbiosis".

Chapter 11

Scheme: 5D Double Helix and Dimensional Solutions

11.1 Physical 5-Dimensions

Nodes, Topology ($\mathbf{A}$), Constraints (C), State Transitions, and State Vectors ($\mathbf{x}(t)$).

11.2 Parallel Universe Simulation

Simulating N future trajectories in compute space before physical execution:

$$\mathbf{x}_k(t + \Delta t) = \mathcal{F}(\mathbf{x}(t), \mathbf{u}_k(t)), \quad k = 1, 2, \dots, N \quad (11.1)$$

Chapter 12

Capability: Rigid Manifold $\Pi_{\perp}$ Normal Degree-of-Freedom Stripping

Normal degree-of-freedom stripping:

$$\mathbf{x}_{\perp}(t) = \mathbf{x}(t) - \Pi_{\perp} \cdot \mathbf{x}(t) \longrightarrow 0 \quad (12.1)$$

Chapter 13

Mechanism: Control Domain Dimension $\dim(C)$ and Mismatch Angle θ Elimination

Decision writeback: $\dim(C) \leq \dim(O)$. Automation channels write control vectors $\mathbf{u}(t)$ back directly into physical nodes.

Chapter 14

Route: Micro Perception to Macro Control

Micro perception filtering $\Delta(t) = \mathbf{\Pi} \cdot (\mathbf{x}_{\text{real}}(t) - \mathbf{x}_{\text{model}}(t)) \rightarrow$ Compute space pruning $\rightarrow$ Macro writeback.

Chapter 15

Work: Landauer Principle and Info-Energy Conservation

$$\Delta E \geq k_B T \ln 2 \cdot \Delta I \tag{15.1}$$

Chapter 16

Evolution: Passive Residual Triggering and Paradigm Shifts

Residual triggering $\Delta > \theta_{\text{trigger}} \implies \Phi : \Pi_k \rightarrow \Pi_{k+1}$.

Part IV

Volume IV: Cross-Domain Empirical Proof and Symbiotic Civilization

Chapter 17

Cross-Domain Dimensionality Reduction and Decisive Verification

Empirical verification across three grand battlefields:

1. **AlphaFold 2 Protein Folding:** Resolving Levinthal Paradox by clipping phase space via spatial boundary operator Π .
2. **Holographic Principle and Black Hole Thermodynamics:** Bekenstein-Hawking entropy $S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}$ demonstrating horizon boundary definition.
3. **Goodhart 2008 Financial Collapse:** Subprime mortgage crisis resulting from unclipped heavy-tailed distributions lacking operator $\mathbf{E}_{\text{sp}}$.

Chapter 18

Ultimate Symbiosis: Carbon-Based Prior and Silicon-Based Compute

18.1 Four-Group Empowerment in the Era of LLMs and Multi-Agent Systems

1. **Top Executives:** Parallel universe simulation and zero-cost risk pre-computation.
2. **System Engineers:** $O(N!)$ matrix solving and auto-writeback into physical actuators.
3. **Knowledge Workers:** Sub-second residual extraction and high-precision noise filtering.
4. **General Public:** Natural language intuitive control converting intent into lower-level control vectors $\mathbf{u}(t)$.

Epilogue: To Future Seekers of Order

Order is neither a static entity nor an arbitrary construct. It is the eternal physics of observer boundary definition. We leave this physics constitution to all future explorers who seek order amidst chaos.